An Elementary Proof of Robust Bivariate Reed–Solomon Gluing

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Abstract

We give a self-contained proof of the robust bivariate gluing theorem underlying the Polishchuk–Spielman low-degree test and the Reed–Solomon PCP of proximity of Ben-Sasson and Sudan. The proof uses only Lagrange interpolation, greatest common divisors, elementary linear algebra, determinants, and univariate root counting. Its central device is a low-bidegree polynomial that vanishes on all row–column disagreements. Multiplying by this locator makes the local views exactly compatible; a padded Sylvester determinant then forces the locator to divide the resulting bivariate polynomial globally. The remaining errors are confined to a small rectangle of exceptional rows and columns.

Proof in five moves. Given low-degree row and column polynomials that disagree on a sparse set:

$$\text{locator} \longrightarrow \text{exact interpolation} \longrightarrow \text{determinant divisibility},$$

$$\text{exceptional fibers} \longrightarrow \text{robust testing}.$$

The only point requiring care is the multiplicity with which a good fiber is a root of the determinant. It follows from a one-line rank-deficiency lemma.

1 The gluing problem

Let $X, Z \subseteq \mathbb{F}$ be finite sets of distinct field elements, with

$$|X| = n_X, \qquad |Z| = n_Z.$$

For each $z \in Z$, suppose we are given a column polynomial

$$r_z(U) \in \mathbb{F}[U], \qquad \deg r_z \le d_X,$$

and for each $x \in X$, a row polynomial

$$c_x(V) \in \mathbb{F}[V], \qquad \deg c_x \le d_Z.$$

At a grid point (x, z), the two local views propose the values $r_z(x)$ and $c_x(z)$. Their disagreement set is

$$S := \{(x, z) \in X \times Z : r_z(x) \neq c_x(z)\}, \qquad s := |S|.$$

The aim is to prove that, if S is sparse and the two degree bounds leave enough slack, then almost all local views come from one polynomial

$$Q(U, V), \qquad \deg_U Q \le d_X, \quad \deg_V Q \le d_Z.$$

The robust conclusion will be pointwise: outside the original disagreement set and a small exceptional rectangle, both local views equal Q.

2 Two elementary preliminaries

2.1 Exact tensor interpolation

Lemma 2.1 (Tensor interpolation)**.** *Let $H : X \times Z \to \mathbb{F}$. Suppose every column $H(\cdot, z)$ is represented on X by a polynomial of degree at most $D_X < n_X$, and every row $H(x, \cdot)$ is represented on Z by a polynomial of degree at most $D_Z < n_Z$. Then there is a unique polynomial $P(U, V)$ with*

$$\deg_U P \le D_X, \qquad \deg_V P \le D_Z,$$

such that

$$P(x, z) = H(x, z) \qquad (x \in X, z \in Z).$$

Moreover, for each fixed z, $P(U, z)$ is the prescribed column polynomial, and for each fixed x, $P(x, V)$ is the prescribed row polynomial.

Proof. Choose distinct points $x_0, \dots, x_{D_X} \in X$, and let $L_0(U), \dots, L_{D_X}(U)$ be their Lagrange basis polynomials. Let $h_i(V)$ be the degree-$\le D_Z$ polynomial representing the row through x_i. Set

$$P(U, V) := \sum_{i=0}^{D_X} L_i(U) h_i(V).$$

Then P has the stated bidegree. For fixed z, the polynomial $P(U, z)$ agrees with the prescribed column polynomial at $D_X + 1$ points and both have degree at most D_X, so they are equal. The row assertion and uniqueness follow immediately. □

2.2 Rank deficiency gives root multiplicity

Lemma 2.2 (Determinant multiplicity)**.** *Let $M(T) \in \mathbb{F}[T]^{N \times N}$, let $t \in \mathbb{F}$, and suppose*

$$\operatorname{rank} M(t) \le N - r.$$

Then

$$(T - t)^r \mid \det M(T).$$

Proof. By constant invertible row and column operations, put

$$M(t) = \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix}, \qquad q \le N - r.$$

These operations change the determinant only by a nonzero constant. In the resulting polynomial matrix, every entry outside the upper-left $q \times q$ block is divisible by $T - t$. A monomial in the determinant expansion can use at most q entries from that block; hence it uses at least $N - q \geq r$ entries divisible by $T - t$. Every determinant monomial is therefore divisible by $(T - t)^r$. $\square$

3 The elementary divisibility lemma

This is the algebraic heart of the proof. It is the corrected form of the Polishchuk–Spielman divisibility principle: the fiberwise quotients must have the degree predicted by the global degree gap.

Theorem 3.1 (Fiberwise divisibility)**.** *Let $A, B \in \mathbb{F}[U, V]$, with $A \neq 0$. Let*

$$a_X \leq b_X, \qquad a_Z \leq b_Z$$

be nonnegative integers such that

$$\deg_U A \leq a_X, \quad \deg_V A \leq a_Z, \quad \deg_U B \leq b_X, \quad \deg_V B \leq b_Z.$$

Write

$$d_X := b_X - a_X, \qquad d_Z := b_Z - a_Z.$$

Suppose there are sets $X_0, Z_0 \subseteq \mathbb{F}$ of distinct elements, of sizes m_X, m_Z, such that:

- *for every $x \in X_0$,*

$$B(x, V) = p_x(V)A(x, V) \quad \text{for some } p_x \in \mathbb{F}[V] \text{ with } \deg p_x \leq d_Z;$$

- *for every $z \in Z_0$,*

$$B(U, z) = q_z(U)A(U, z) \quad \text{for some } q_z \in \mathbb{F}[U] \text{ with } \deg q_z \leq d_X.$$

If

$$\frac{b_X}{m_X} + \frac{b_Z}{m_Z} < 1, \tag{1}$$

then $A \mid B$ in $\mathbb{F}[U, V]$. Writing $B = AQ$, one has

$$\deg_U Q \leq d_X, \qquad \deg_V Q \leq d_Z.$$

Proof. The case $B = 0$ is immediate, so assume $B \neq 0$.

1. Normalize the degree bounds. If neither A nor B attains its stated U-degree bound, decrease both a_X and b_X by one. This preserves d_X, preserves every hypothesis, and only decreases the left side of (1). Repeat until

$$\deg_U A = a_X \quad \text{or} \quad \deg_U B = b_X. \tag{2}$$

Do the same in the V-variable. Thus, in each variable, at least one polynomial attains its bound.

2. Divide out the gcd. Let $G := \gcd(A, B)$, and write

$$g_X := \deg_U G, \qquad g_Z := \deg_V G.$$

Replace A, B by $A/G, B/G$, and replace the four degree bounds by

$$a_X - g_X, \quad b_X - g_X, \quad a_Z - g_Z, \quad b_Z - g_Z.$$

The gaps d_X, d_Z do not change. At most g_X values $x \in X_0$ satisfy $G(x, V) \equiv 0$, since each such value contributes a factor $U - x$ to G. On every other good fiber we may cancel $G(x, V)$. Thus the new number of good U-fibers is at least $m_X - g_X$, and similarly the new number of good V-fibers is at least $m_Z - g_Z$.

Since (1) implies $b_X < m_X$ and $b_Z < m_Z$,

$$\frac{b_X - g_X}{m_X - g_X} \le \frac{b_X}{m_X}, \qquad \frac{b_Z - g_Z}{m_Z - g_Z} \le \frac{b_Z}{m_Z}.$$

Hence the rate condition survives. The equalities in (2) and its V-analogue also survive, because degrees are additive under multiplication.

It therefore suffices to prove the theorem when

$$\gcd(A, B) = 1.$$

We will show that then A is constant.

3. Form the coefficient determinant in the U-variable. Work over the field $K := \mathbb{F}(V)$. Consider the K-linear map

$$\Phi_U : K[U]_{<a_X} \oplus K[U]_{<b_X} \longrightarrow K[U]_{<a_X+b_X}, \qquad (R, S) \longmapsto RB + SA.$$

The source and target have the same dimension $a_X + b_X$. The map is injective. Indeed, if

$$RB + SA = 0,$$

then coprimality implies $A \mid R$ and $B \mid S$. If $\deg_U A = a_X$, the inequality $\deg R < a_X$ forces $R = 0$, and then $S = 0$. If instead $\deg_U B = b_X$, the same argument begins with $S = 0$.

Let $M_U(V)$ be the matrix of Φ_U in monomial bases, and put

$$D_U(V) := \det M_U(V).$$

Thus D_U is a nonzero polynomial. The matrix consists of a_X shifted coefficient rows of B and b_X shifted coefficient rows of A. Consequently

$$\deg_V D_U \le a_X b_Z + a_Z b_X =: D. \tag{3}$$

4. Every good V-fiber is a high-multiplicity root. Fix $z \in Z_0$. Since

$$B(U, z) = q_z(U) A(U, z), \qquad \deg q_z \le b_X - a_X,$$

for every $0 \le i < a_X$ the polynomial $U^i B(U,z)$ is a linear combination of

$$A(U,z), UA(U,z), \dots, U^{b_X-1}A(U,z).$$

Thus the row space of $M_U(z)$ is spanned by its b_X rows coming from A, and

$$\operatorname{rank} M_U(z) \le b_X.$$

The matrix has size $a_X + b_X$, so Theorem 2.2 gives

$$(V-z)^{a_X} \mid D_U(V).$$

The elements of Z_0 are distinct, hence

$$a_X m_Z \le \deg_V D_U \le D. \tag{4}$$

Repeating the entire construction with U and V interchanged gives

$$a_Z m_X \le D. \tag{5}$$

5. The numerical contradiction. Using (4) and (5),

$$\begin{aligned} D &= a_X b_Z + a_Z b_X \\ &= \frac{b_Z}{m_Z}(a_X m_Z) + \frac{b_X}{m_X}(a_Z m_X) \\ &\le \left(\frac{b_X}{m_X} + \frac{b_Z}{m_Z}\right) D. \end{aligned}$$

By (1), the coefficient is strictly smaller than one. Hence $D = 0$. Then (4) and (5) imply

$$a_X = a_Z = 0.$$

Thus the coprime reduction of A is constant, so the original A divides the original B.

Finally, in the coprime reduction the quotient has U-degree at most $b_X = d_X$ and V-degree at most $b_Z = d_Z$. The quotient and the degree gaps were unchanged by all reductions, so the same bounds hold for the original quotient. □

Remark 3.2 (Why the quotient-degree condition matters)**.** The rank bound on $M_U(z)$ uses

$$\deg q_z \le b_X - a_X.$$

Without it, the shifted copies $U^i B(U,z)$ need not lie in the span of the b_X shifted copies of $A(U,z)$, and the determinant need not vanish with multiplicity a_X. This is the subtle point missing from the original formulation of the divisibility lemma and repaired in the corrected proof of Ben-Sasson, Carmon, Ishai, Kopparty, and Saraf.

4 Robust gluing

We now apply Theorem 3.1 to the row and column polynomials from Section 1. For readability, the theorem is stated with arbitrary locator degrees; the familiar δ-form follows immediately.

Theorem 4.1 (Robust bivariate gluing)**.** *Let $(r_z)_{z\in Z}$ and $(c_x)_{x\in X}$ be as in Section 1, with disagreement set S of size s. Suppose there are nonnegative integers ℓ_X, ℓ_Z such that*

$$(\ell_X + 1)(\ell_Z + 1) > s \tag{6}$$

and

$$\frac{\ell_X + d_X}{n_X} + \frac{\ell_Z + d_Z}{n_Z} < 1. \tag{7}$$

Then there is a polynomial $Q(U, V)$ satisfying

$$\deg_U Q \le d_X, \qquad \deg_V Q \le d_Z,$$

such that

$$|\{(x, z) : r_z(x) \ne Q(x, z) \text{ or } c_x(z) \ne Q(x, z)\}| \le s + \ell_X \ell_Z. \tag{8}$$

Proof. **1. Choose a locator.** The vector space of polynomials of bidegree at most (ℓ_X, ℓ_Z) has dimension

$$(\ell_X + 1)(\ell_Z + 1) > s.$$

The conditions of vanishing at the s points of S are homogeneous linear equations, so there is a nonzero polynomial $L(U, V)$ with

$$\deg_U L \le \ell_X, \qquad \deg_V L \le \ell_Z,$$

such that

$$L(x, z) = 0 \qquad ((x, z) \in S).$$

2. Multiply away the disagreements. Define

$$H(x, z) := L(x, z) r_z(x).$$

If $(x, z) \notin S$, then $r_z(x) = c_x(z)$; if $(x, z) \in S$, then $L(x, z) = 0$. Hence, at every grid point,

$$H(x, z) = L(x, z) c_x(z)$$

as well.

Every column of H is represented by $L(U, z) r_z(U)$, of degree at most $\ell_X + d_X$, and every row is represented by $L(x, V) c_x(V)$, of degree at most $\ell_Z + d_Z$. Condition (7) implies that these two degree bounds are strictly below n_X, n_Z. By Theorem 2.1, there is a polynomial $P(U, V)$ of bidegree at most

$$(\ell_X + d_X, \ell_Z + d_Z)$$

satisfying the formal fiber identities

$$P(U, z) = L(U, z) r_z(U), \qquad P(x, V) = L(x, V) c_x(V). \tag{9}$$

3. Divide globally. Apply Theorem 3.1 to $A = L$ and $B = P$, with

$$a_X = \ell_X, \quad b_X = \ell_X + d_X, \qquad a_Z = \ell_Z, \quad b_Z = \ell_Z + d_Z.$$

The fiber quotients in (9) have exactly the required degree bounds, and (7) is the rate condition. Therefore

$$P = LQ$$

for a polynomial Q of bidegree at most (d_X, d_Z).

4. Count exceptional fibers. For every $z \in Z$,

$$L(U, z)(r_z(U) - Q(U, z)) = 0.$$

Since $\mathbb{F}[U]$ is an integral domain, either $r_z(U) = Q(U, z)$ or $L(U, z) \equiv 0$. The second alternative can occur for at most ℓ_Z values of z, because each such value contributes a distinct factor $V - z$ to L. Thus there are at most ℓ_Z bad columns. Symmetrically, there are at most ℓ_X bad rows.

Outside S, the row and column values coincide. Hence a point outside S where either local view differs from Q must lie simultaneously on a bad row and a bad column. There are at most $\ell_X \ell_Z$ such points, proving (8). □

Corollary 4.2 (The usual δ-form)**.** *Suppose* $0 < \delta < 1$,

$$\frac{s}{n_X n_Z} < \delta^2,$$

and

$$\frac{d_X}{n_X} + \frac{d_Z}{n_Z} + 2\delta < 1. \tag{10}$$

Then there is a polynomial Q *of bidegree at most* (d_X, d_Z) *for which*

$$\Pr_{(x,z)\in X\times Z}[r_z(x) \neq Q(x, z) \text{ or } c_x(z) \neq Q(x, z)] < 2\delta^2.$$

Proof. Take

$$\ell_X := \lfloor \delta n_X \rfloor, \qquad \ell_Z := \lfloor \delta n_Z \rfloor.$$

Then

$$(\ell_X + 1)(\ell_Z + 1) > \delta^2 n_X n_Z > s, \qquad \ell_X \ell_Z \leq \delta^2 n_X n_Z,$$

and (10) implies (7). Apply Theorem 4.1. □

5 The robust row–column test

The structural theorem immediately gives the robust low-degree test used in the Reed–Solomon PCPP.

Let

$$\mathcal{C}_X := \mathrm{RS}_X(d_X) := \{(p(x))_{x\in X} : \deg p \leq d_X\},$$

and define $\mathcal{C}_Z := \mathrm{RS}_Z(d_Z)$ similarly. Their tensor product $\mathcal{T} := \mathcal{C}_X \otimes \mathcal{C}_Z$ is the set of evaluations on $X \times Z$ of polynomials of bidegree at most (d_X, d_Z).

For a table $f : X \times Z \to \mathbb{F}$, define the average column and row distances

$$\varepsilon_X(f) := \mathbb{E}_{z \in Z}\, \Delta(f(\cdot, z), \mathcal{C}_X), \qquad \varepsilon_Z(f) := \mathbb{E}_{x \in X}\, \Delta(f(x, \cdot), \mathcal{C}_Z),$$

where line distances are normalized by the line length. Put

$$\gamma := 1 - \frac{d_X}{n_X} - \frac{d_Z}{n_Z}.$$

Corollary 5.1 (Robust row–column test)**.** *If $\gamma > 0$, then every table $f : X \times Z \to \mathbb{F}$ satisfies*

$$\Delta(f, \mathcal{T}) \le \frac{4}{\gamma^2}(\varepsilon_X(f) + \varepsilon_Z(f)). \tag{11}$$

Equivalently, the tester that chooses a uniformly random row or column has expected line distance at least

$$\frac{\gamma^2}{8}\, \Delta(f, \mathcal{T}).$$

Proof. For each column choose a nearest polynomial r_z of degree at most d_X, and for each row choose a nearest polynomial c_x of degree at most d_Z. Let R, C be the resulting two tables and put

$$\eta := \varepsilon_X(f) + \varepsilon_Z(f).$$

At every point where $R \neq C$, at least one of them differs from f, so

$$\Delta(R, C) \le \eta.$$

First suppose $\eta < \gamma^2/4$. Choose δ with

$$\sqrt{\eta} < \delta < \gamma/2.$$

Then Theorem 4.2 gives $Q \in \mathcal{T}$ such that the union of the mismatch sets $R \neq Q$ and $C \neq Q$ has density below $2\delta^2$. Therefore

$$\begin{aligned} \Delta(f, Q) &\le \min\{\Delta(f, R) + \Delta(R, Q),\ \Delta(f, C) + \Delta(C, Q)\} \\ &\le \frac{\eta}{2} + 2\delta^2. \end{aligned}$$

Letting $\delta \downarrow \sqrt{\eta}$ gives

$$\Delta(f, \mathcal{T}) \le \frac{5}{2}\eta.$$

Since $\gamma \le 1$, this is at most $4\eta/\gamma^2$.

If instead $\eta \ge \gamma^2/4$, the trivial bound $\Delta(f, \mathcal{T}) \le 1$ gives

$$\Delta(f, \mathcal{T}) \le \frac{4}{\gamma^2}\eta.$$

This proves (11). The equivalent tester statement follows because the expected line distance is $(\varepsilon_X + \varepsilon_Z)/2$. □

6 What remains of the Polishchuk–Spielman argument

The proof above is elementary, but it is not a different phenomenon. It is a stripped-down presentation of the robust gluing mechanism behind the Polishchuk–Spielman bivariate low-degree test:

1. A sparse disagreement set admits a low-bidegree locator.
2. Multiplication by the locator makes all row and column descriptions exactly consistent.
3. Exact consistency gives one bivariate polynomial by interpolation.
4. A coefficient determinant forces fiberwise divisibility to be global.
5. Only fibers on which the locator vanishes identically can remain exceptional.

The determinant $D_U(V)$ is the padded Sylvester determinant, but no theorem about resultants is needed: its nonvanishing follows from injectivity of a multiplication map, its degree is read directly from its rows, and its root multiplicities follow from Theorem 2.2.

For the Reed–Solomon PCP of proximity, Theorem 5.1 is the needed soundness input: if a bivariate table is globally far from the tensor Reed–Solomon code, then a random row or column has noticeable distance from its corresponding one-dimensional code. The recursive degree-splitting construction and its proof-length bookkeeping are separate from this gluing lemma.

References

[1] A. Polishchuk and D. A. Spielman. Nearly-linear size holographic proofs. In *Proceedings of the 26th Annual ACM Symposium on Theory of Computing*, pages 194–203, 1994.

[2] E. Ben-Sasson and M. Sudan. Short PCPs with polylog query complexity. *SIAM Journal on Computing*, 38(2):551–607, 2008.

[3] E. Ben-Sasson, D. Carmon, Y. Ishai, S. Kopparty, and S. Saraf. Proximity gaps for Reed–Solomon codes. ECCC Report TR20-083, Revision 3, 2021. Appendix D gives the corrected Polishchuk–Spielman divisibility lemma.